Information Fusion 134 (2026) 104368



Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus





Modality-aware fusion with alignment and balancing for robust multimodal emotion recognition

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ARTICLE INFO

Keywords:
Multimodal learning
Modality distribution gap
Modality imbalance
Dual-level alignment
Modality-adaptive re-initialization

ABSTRACT

Multimodal learning has demonstrated great potential in affective recognition by integrating complementary information from diverse modalities. However, inherent differences in data sources, representational structures, and semantic characteristics across modalities often give rise to two major challenges: modality distribution gap and modality imbalance. Existing approaches typically focus on only one of these issues and lack a unified solution. Particularly, dominant strategies for addressing modality imbalance often rely on enhancing weak modalities or suppressing strong ones. Such approaches risk disrupting the original learning dynamics of the model, potentially leading to a degradation in performance. To this end, we propose a unified alignment and re-initialization framework, MAAB-ER, designed to mitigate distributional divergence and representation imbalance across modalities jointly. The framework consists of two key modules: (1) A dual-level alignment module that combines distribution-level and modality-level alignment losses to enhance feature consistency and discriminability from both statistical and semantic perspectives; (2) A modality-adaptive re-initialization mechanism that dynamically adjusts each modality's encoder parameters based on its generalization performance, enabling modality-aware reconstruction, strengthening weaker modality representations, and improving overall fusion quality and model robustness. Extensive experiments on the DEAP, SEED-IV, and SEED-V datasets demonstrate that the proposed MAAB-ER framework consistently outperforms several state-of-the-art methods in terms of alignment and imbalance, validating its effectiveness and general applicability in multimodal emotion recognition (MER) tasks.

1. Introduction

Emotion is an essential part of our lives, profoundly influencing our cognition, behavior, and social interactions. Human emotions can be expressed through multiple modalities, and leveraging diverse sources of information to infer and recognize these emotions has become increasingly important in fields such as human-computer interaction (HCI) [1], affective computing (AC) [2], and psychological assessment [3]. Among various modalities, electroencephalogram (EEG) and video data are widely used in emotion recognition research [4,5]. EEG signals provide a more direct and objective measure of the neural activities underlying emotional states; however, they are often susceptible to noise and substantial inter-subject variability. On the other hand, video data captures external cues such as facial expressions and eye movements, which are rich in emotional information but can be influenced by environmental factors including lighting conditions and subjective interpretations. Given the complementary nature and limitations of these two modalities, effectively integrating EEG and video data presents a promising and critical avenue to advance multimodal emotion recognition (MER) performance.

However, as illustrated in Fig. 1(a), multimodal data from heterogeneous sources often show unequal contributions, inconsistent signal quality, and heterogeneous discriminative capacities, giving rise to the problem of modality imbalance. This refers to the phenomenon where stronger modalities dominate the model optimization while weaker modalities contribute marginally, ultimately affecting the stability and effectiveness of fusion [6,7]. Recent studies have further shown that certain modalities can systematically overshadow others during training, thereby exacerbating imbalance and, in some cases, even causing multimodal models to perform worse than their single-modality counterparts

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https://doi.org/10.1016/j.inffus.2026.104368
Received 22 July 2025; Received in revised form 5 April 2026; Accepted 7 April 2026
Available online 12 April 2026
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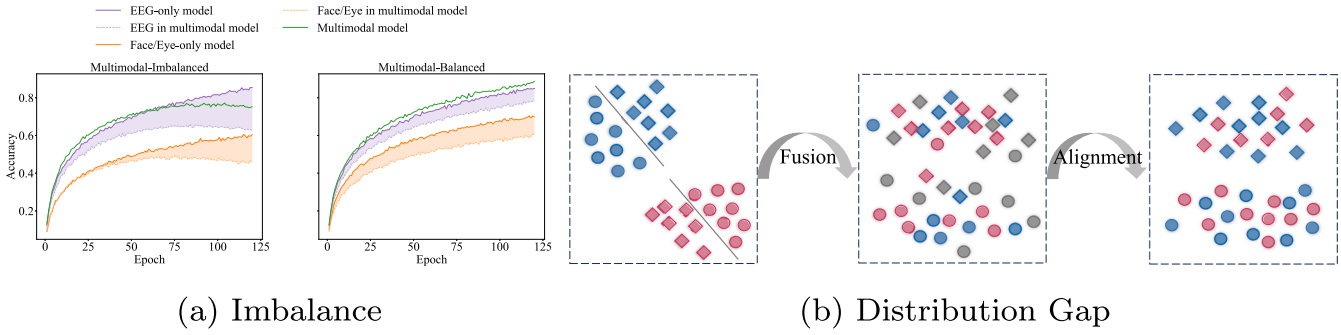


Fig. 1. (a) Illustration of modality imbalance, where conventional multimodal methods may yield inferior performance compared to certain unimodal baselines and even degrade the performance of stronger modalities. By introducing a modality balancing mechanism, the performance across all modalities can be improved. (b) Illustration of multimodal alignment. Without proper alignment, multimodal fusion may introduce redundancy and noise, weakening the representation quality. With alignment, features from different modalities are better aligned in the semantic space, enabling more effective fusion and improved discriminative ability.

[8]. To mitigate this issue, various approaches have been proposed. A common strategy involves suppressing the influence of dominant (well-performing) modalities while emphasizing the training of weaker (underperforming) ones, to achieve a more balanced and comprehensive representation [9–11]. However, such approaches often neglect the intrinsic properties of individual modalities. Specifically, some modalities may inherently lack discriminative, label-relevant information or may be contaminated with substantial noise. In these cases, their poor performance is not necessarily due to insufficient training, but rather stems from their limited representational capacity. As a result, excessively focusing on such inherently weak modalities can introduce additional noise and irrelevant features, ultimately degrading the overall predictive performance of the multimodal model. Therefore, designing a principled and effective balancing mechanism remains a challenge.

In addition to imbalance, Fig. 1(b) highlights another fundamental challenge, referred to as modality distribution discrepancy. This issue stems from the intrinsic heterogeneity across modalities with respect to their temporal dynamics, spatial structures, feature distributions, and semantic expressiveness. Such discrepancies can lead to misalignment during feature extraction and fusion, causing redundancy, feature conflicts, or the loss of complementary emotional cues [12]. Although several studies have attempted to address this issue using techniques such as cross-modal attention [13], contrastive learning [14], or shared latent space projection [15], achieving robust and scalable alignment across varying contexts and individual differences remains an open challenge. Moreover, in EEG-based multimodal frameworks that incorporate facial or eye-movement signals [16], modalities differ markedly in temporal resolution, spatial topology, and semantic representation. These intrinsic discrepancies further exacerbate alignment difficulty. Therefore, designing adaptive and modality-aware alignment mechanisms that can bridge semantic and structural gaps between modalities remains a pressing and unresolved challenge in MER.

To address the challenges of modality misalignment and imbalance in MER, this paper proposes MAAB-ER, a unified framework that integrates alignment and balancing mechanisms in fusion and training stages. Firstly, the modality-specific encoders extract discriminative features and are then refined using a combination of self-attention and cross-modal Mamba modules to capture intra- and inter-modal dependencies. Additionally, an attention-based fusion strategy integrates these features into a shared representation. To reduce modality gaps, we introduce a dual-level alignment loss: a distribution-level contrastive loss aligns feature distributions in high-dimensional space, while a modality-level alignment loss promotes semantic consistency between modality-specific and cross-modal features. To address modality imbalance, we propose a modality-adaptive re-initialization strategy guided by each modality's learning status, which aggressively updates well-learned modalities to prevent overfitting while conservatively adjusting underperforming ones to retain informative features and suppress noise. This adaptive re-initialization mechanism enables the model to retain valuable knowledge while rebalancing learning dynamics across modalities, ultimately enhancing generalization and achieving effective multimodal fusion.

The paper makes the following key contributions:

1. **Dual-level alignment mechanism:** We combine distribution-level contrastive learning for coarse-grained alignment with modality-aware contrastive learning for fine-grained alignment, effectively enhancing cross-modal representation consistency while accommodating the semantic ambiguity of emotions.
2. **Modality-adaptive re-initialization:** We introduce a strategy that dynamically calibrates each modality-specific encoder based on its learning status, reducing overreliance on dominant modalities, mitigating noise from weaker modalities, and preserving informative prior knowledge to improve robustness and generalization.
3. **Extensive empirical validation:** We evaluate MAAB-ER on three benchmark emotion recognition datasets. Results show that our method consistently outperforms existing alignment- and perception-based baselines, demonstrating the effectiveness of both the alignment and modality-adaptive mechanisms in enhancing overall model performance.

2. Related work

2.1. Modality alignment approaches for MER

Modality alignment aims to reduce representational discrepancies across heterogeneous modalities, enabling coherent fusion and improved emotional inference. Existing approaches can be broadly grouped into distribution-level, instance-level, and structure-aware alignment strategies. For **multi-level and distribution-level alignment**, several works focus on aligning modality distributions at different granularity levels. Wang et al. [17] introduced the Multi-Granularity Cross-Modal Alignment (MGCMA), which jointly performs distribution-level, instance-level, and token-level alignment to overcome the limitations of single-granularity strategies. Jimenez et al. [18] proposed a hierarchical alignment paradigm that performs coarse global alignment followed by fine-grained domain-specific alignment to achieve robust semantic consistency across modalities. For **instance-level and sample-level alignment**, Wang et al. [19] developed a framework that performs inter-modality and intra-sample alignment, reducing modality discrepancies by aligning sample-level representations. Zhang et al. [20] performed instance-level and prototype-level contrastive alignment to structure multimodal features more coherently in latent space. For **structure-aware or disentangled alignment**, more specialized alignment techniques target the structural decomposition of modalities. Meng et al. [21] introduced a Masked Graph Learning with Recursive

Alignment (MGLRA) strategy, aligning modalities before fusion based on graph-structured representations. Sun et al. [22] proposed a fine-grained disentangled representation learning approach that separates modality-shared and modality-private components to minimize distributional inconsistency and redundancy.

Although previous alignment methods have demonstrated strong performance in commonly studied modality pairs, such as audio-text or audio-video, they often focus on scenarios where data are more homogeneous and large-scale datasets are readily available. Such setups simplify distribution matching and reduce cross-modal discrepancies. In contrast, EEG signals present unique challenges, including high inter-subject variability, low signal-to-noise ratio, and fine-grained temporal dynamics, while facial and eye-tracking modalities differ in spatial structure, semantic representation, and temporal resolution. Therefore, the combination of EEG with facial or eye-tracking data introduces complex heterogeneity that is not fully addressed by existing alignment strategies. Motivated by this gap, our work focuses on EEG combined with facial or eye-tracking modalities to tackle a practically relevant and underexplored scenario. To bridge these heterogeneous representations, we propose a dual-level alignment strategy that incorporates both distribution-level and modality-level constraints, enabling more robust and semantically consistent feature integration across these challenging modalities.

2.2. Modality balancing approaches for MER

Modality balancing aims to regulate the learning dynamics of different modalities so that no single modality disproportionately dominates optimization. Current approaches generally fall into sample-adaptive weighting, gradient modulation, and prototype-level or representation-level balancing. For **sample-adaptive or hierarchical weighting**, Fang et al. [23] proposed a mixture-of-experts strategy that dynamically allocates modality importance at the sample level, integrating modality-invariant and modality-specific cues. Nguyen et al. [24] introduced Adaptive Feature Weighting (AFW) and Adaptive Modality Weighting (AMW), which adjust feature- and modality-level importance based on intra- and inter-modal interactions. For **gradient-based balancing**, Peng et al. [8] developed an on-the-fly gradient modulation strategy that monitors gradient discrepancies across modalities and adaptively adjusts optimization processes. Zhang et al. [25] minimized a surrogate alignment loss to automatically regulate modality weights, achieving balanced domain alignment during training. For **prototype-driven or representation-level balancing**, Fan et al. [11] introduced the Prototypical Modality Rebalance (PMR) method, which selectively stimulates underperforming modalities while reducing interference from others. Meng et al. [26] proposed a multimodal adversarial learning framework that enhances representation boundaries and indirectly mitigates class imbalance, contributing to more stable multimodal learning.

Despite these advances, most balancing strategies adopt a “suppress strong, emphasize weak” paradigm, assuming weaker modalities underperform due to insufficient optimization. In practice, some modalities are inherently noisy or contain limited label-relevant information. Blindly boosting them may introduce additional noise and degrade overall performance. This limitation motivates our adaptive, informativeness-aware re-initialization strategy, which balances modalities according to their true discriminative value rather than only their learning status.

2.3. Cross-modal fusion approaches for MER

Cross-modal fusion plays a crucial role in MER, as it enables complementary information exchange across different data sources such as speech, text, vision, and physiological signals. For **transformer-based fusion**, many studies rely on cross-attention mechanisms to capture inter-modal dependencies. Khan et al. [27] proposed MemoCMT, which employs a cross-modal Transformer to analyze local and global correlations between speech and text jointly. Praveen et al. [28] introduced the Incongruity-Aware Cross-Attention (IACA), which incorporates a two-stage gating mechanism to select semantically relevant features across modalities adaptively. Liu et al. [29] developed the TACFN framework, in which each modality first performs intra-modal feature selection through self-attention before interacting with others, thereby enabling more informative and efficient cross-modal feature exchange.

Praveen et al. [30] further proposed the Recursive Joint Cross-Modal Attention (RJCMA), which recursively models both intra-modal and inter-modal relationships among audio, visual, and text signals. For **SSM- and Mamba-based fusion**, recent works explore state-space models as computationally efficient alternatives to Transformers due to their linear complexity and strong ability to model long-range dependencies. Ping et al. [31] proposed MambaMER, which introduces Mutual-Cross-Mamba and Merge-Mamba modules to capture long-range cross-modal interactions effectively. Liu et al. [32] designed the CrossMamba module, replacing the Transformer’s quadratic attention with a linear-complexity state-space mechanism for efficient multimodal fusion. Li et al. [33] developed a Coupled SSM model that maintains independent intra-modal state processes while coupling cross-modal state transitions, enabling coordinated yet modality-aware fusion.

Although these methods improve efficiency and capture global dependencies, many fusion mechanisms remain unidirectional or sequential, limiting their ability to exploit complementary multimodal information, particularly when modalities have heterogeneous temporal resolutions or structural characteristics. This limitation motivates our bidirectional cross-fusion Mamba mechanism, which establishes two-way semantic interactions while preserving efficient long-range modeling, enabling more robust and balanced multimodal fusion.

3. Method

The proposed MAAB-ER model, as illustrated in Fig. 2, is designed around two key components: modality alignment and modality balancing. First, the model performs feature extraction on each input modality to obtain modality-specific embeddings. During the fusion process, a dual-level modality alignment mechanism is employed to reduce cross-modal discrepancies at the distribution-level and modality-level. Next, a modality-aware balancing mechanism is introduced to re-evaluate the learning status of each modality.

3.1. Modality-specific feature extract

Due to the distinct feature characteristics of each modality, using a shared feature extractor is often insufficient to capture modality-specific representations effectively. To address this issue, we employ dedicated feature encoders for each modality to better extract discriminative and modality-aware information. These specific modality encoders are based on lightweight convolutional neural networks rather than directly adopting existing backbone models, as illustrated in Table 1. Specifically, we assume that the shallow features of each modality are first extracted and denoted as $\mathbf{X}_m \in \mathbb{R}^{N \times L_m}$, where N is the number of data samples and L_m represents the feature dimension of modality

Table 1
Architecture of modality-specific encoders.

Modality	Layer	Type	Kernel	Stride	Channels	Norm	Act
EEG	Conv1	Conv2D	3×3	1×1	$L_m \rightarrow 64$	BN	ReLU
	Conv2	Conv2D	3×3	1×1	$64 \rightarrow 128$	BN	ReLU
	Conv3	Conv2D	3×3	1×1	$128 \rightarrow L$	BN	ReLU
Facial	Conv1	Conv2D	3×3	1×1	$L_m \rightarrow 64$	BatchNorm	ReLU
	Conv2	Conv2D	3×3	1×1	$64 \rightarrow 128$	BatchNorm	ReLU
	Conv3	Conv2D	3×3	1×1	$128 \rightarrow L$	BatchNorm	ReLU
Eye	Conv1	Conv1D	3	1	$L_m \rightarrow 32$	BatchNorm	ReLU
	Conv2	Conv1D	3	1	$32 \rightarrow L$	BatchNorm	ReLU

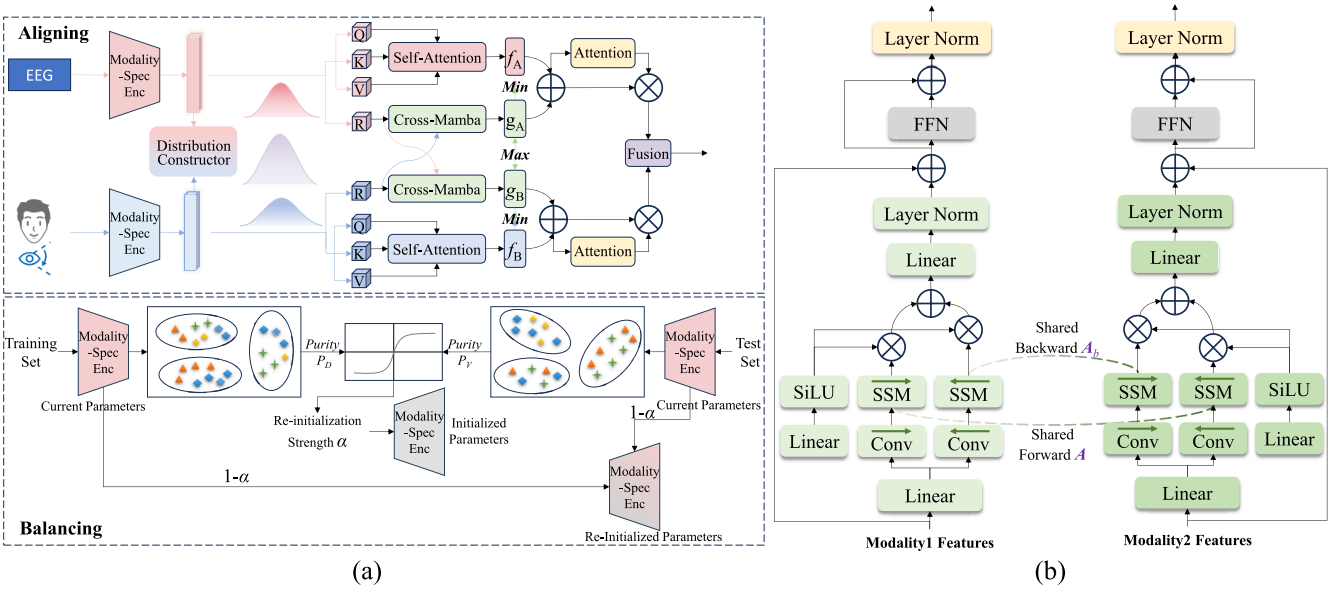


Fig. 2. (a) The overall framework of MAAB-ER, which incorporates dual-level loss and adaptive re-initialization for modality alignment and balancing in MER. (b) The detailed workflow of the Cross-Mamba module within the overall framework, illustrating the process of cross-modal feature interaction and fusion.

m, with corresponding to EEG, facial expression, and eye movement signals, respectively. These shallow features are then passed through modality-specific encoders to obtain enhanced modality-dependent representations. Each modality-specific encoder Enc_m consists of a stack of convolutional layers followed by normalization and non-linear activation functions. Although the encoders follow a unified design principle, their convolutional parameters are modality-dependent. For EEG signals, two-dimensional convolutions are employed to jointly model spatial correlations across channels and temporal dynamics. Facial expression features are processed using spatio-temporal convolution blocks to model local spatial structures and temporal variations. Eye movement features, which are relatively low-dimensional and sparse, are encoded using shallow convolutional layers to preserve fine-grained patterns while reducing the risk of overfitting. For consistency and ease of multimodal fusion, all modality-specific encoders output representations with a unified feature dimension. Formally, the modality-specific representations are obtained as:

$$\mathbf{H}_m = Enc_m(\mathbf{X}_m) \tag{1}$$

where $\mathbf{H}_m \in \mathbb{R}^{N\times L}$ denotes the output representation of modality m, and Enc_m is the corresponding encoder. For consistency and ease of fusion across modalities, each encoder outputs representations with a unified feature dimension L.

3.2. Dual-level modality alignment

3.2.1. Distribution-level alignment

To promote coarse-grained alignment across modalities and reduce distributional discrepancies, we design a distribution-level alignment loss. The objective of this loss is to align the overall feature distributions of different modalities in a shared latent space, thereby facilitating more consistent cross-modal representation learning. Specifically, we denote the two input modalities as **A** and **B** and utilize a distribution encoder to generate multivariate Gaussian distributions $\boldsymbol{z}_A$ and $\boldsymbol{z}_B$ for modality A and modality B, respectively. The distribution encoder is composed of a multi-layer perceptron (MLP) and multi-head attention mechanisms to effectively capture both global and local contextual dependencies within each modality. To enhance the expressive capacity and stabilize the training process, a residual connection with an MLP is introduced after computing the attention-based features. This design allows the encoder to retain essential information from the original feature space while benefiting from the nonlinear transformation of the MLP. Formally, the output distribution is defined as:

$$(\mu, \Sigma) = DisEnc(\mathbf{H}_m) \tag{2}$$

where $\mu \in \mathbb{R}^d$ is the estimated feature mean and $\Sigma \in \mathbb{R}^{d\times d}$ is the positive semi-definite covariance matrix.

After constructing the feature distributions for the two modalities, we implement distribution-level contrastive learning to encourage alignment. Given two multivariate Gaussian distributions $\mathcal{N}_1 \sim (\mu_1, \Sigma_1)$, $\mathcal{N}_2 \sim (\mu_2, \Sigma_2)$, corresponding to encoded features of different modalities, we compute their 2-Wasserstein distance as a metric of distributional discrepancy:

$$\begin{aligned}\mathcal{W}(\mathcal{N}_1, \mathcal{N}_1) &= \|\mu_1 - \mu_2\|_2^2 \\ &\quad + Tr\left(\Sigma_1 + \Sigma_2 - 2\left(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\right)^{1/2}\right) \\ &= \|\mu_1 - \mu_2\|_2^2 + Tr\left(\left(\Sigma_1^{1/2} - \Sigma_2^{1/2}\right)^2\right) \\ &= \|\mu_1 - \mu_2\|_2^2 + \|\sigma_1 - \sigma_2\|_2^2\end{aligned} \tag{3}$$

where $\sigma \in \mathbb{R}^d$ represents the standard deviations of each dimension, and the final simplification assumes that the covariance matrices Σ_1 and Σ_2 are diagonal. This assumption is made for computational efficiency and is reasonable because the encoded features are normalized and approximately uncorrelated across dimensions. Then, we define the similarity between two distributions as:

$$Sim(\mathcal{N}_1, \mathcal{N}_2) = -p \cdot \mathcal{W}(\mathcal{N}_1, \mathcal{N}_1) + b \tag{4}$$

where $(p > 0)$ is a scaling factor and b is the bias term.

Given a batch of K modality pairs $\{(A_i, B_i)\}_{i=1}^K$, we obtain $2K$ distributions, $\mathcal{N}_{Ai} \sim (\mu_{Ai}, \Sigma_{Ai})$, $\mathcal{N}_{Bi} \sim (\mu_{Bi}, \Sigma_{Bi})$. Instances originating from the same modality pair are treated as positive pairs, while all other combinations are considered negative pairs. Based on this setup, the distribution-level contrastive loss is defined as:

$$\mathcal{L}_i^{A2B} = -log\frac{exp(Sim(\mathcal{N}_{Ai}, \mathcal{N}_{Bi})/\tau)}{\sum_{k=1}^K exp(Sim(\mathcal{N}_{Ai}, \mathcal{N}_{Bi})/\tau)} \tag{5}$$

$$\mathcal{L}_i^{B2A} = -log\frac{exp(Sim(\mathcal{N}_{Bi}, \mathcal{N}_{Ai})/\tau)}{\sum_{k=1}^K exp(Sim(\mathcal{N}_{Bi}, \mathcal{N}_{Ai})/\tau)} \tag{6}$$

$$\mathcal{L}_{DA} = \frac{1}{2K}\sum_{i=1}^{K}\left(\mathcal{L}_i^{A2B} + \mathcal{L}_i^{B2A}\right) \tag{7}$$

where τ is a temperature hyperparameter used to scale the logits.

3.2.2. Modality-level alignment

The objective of modality-level alignment is to reduce the representational gap between different modalities by aligning their semantic structures in a shared latent space. To this end, the modality-specific features $\mathbf{H}_A$ and $\mathbf{H}_B$ are first passed through a self-attention module and a cross-Mamba module, which are responsible for extracting modality-specific representations and cross-modal fusion representations, respectively.

In the self-attention module, the features from each modality are linearly projected to obtain the Query (**Q**), Key (**K**), and Value (**V**) matrices:

$$\mathbf{Q}_m = \mathbf{H}_m\mathbf{W}_m^Q, \mathbf{K}_m = \mathbf{H}_m\mathbf{W}_m^K, \mathbf{V}_m = \mathbf{H}_m\mathbf{W}_m^V, \quad m \in \{A, B\} \tag{8}$$

where $\mathbf{W} \in \mathbb{R}^{L\times d}$ is the learnable projection matrices that map the modality-specific hidden representations $\mathbf{H}_m \in \mathbb{R}^{N\times L}$. The self-attention operation for modality $m \in \{A, B\}$ is then computed as:

$$\begin{aligned} f_m^i &= \textit{Self-Att}^i\left(\mathbf{Q}_m^i, \mathbf{K}_m^i, \mathbf{V}_m^i\right) \\ &= Softmax\left(\frac{\mathbf{Q}_m^i\left(\mathbf{K}_m^i\right)^\top}{\sqrt{d_m^i}}\right)\mathbf{V}_m^i, \quad m \in \{A, B\} \end{aligned} \tag{9}$$

where d_m^i denotes the dimension of the query/key vectors in the ith head for modality m. The outputs of all attention heads are concatenated and linearly projected back into the original feature space, after which a residual connection is applied to obtain $\boldsymbol{f}_m$. This residual pathway preserves the original semantic information while enriching the contextual representation through the attention mechanism.

Then, the modality-specific representations $\mathbf{H}_A$ and $\mathbf{H}_B$ are passed through the proposed cross-Mamba module (as shown in Fig. 2(b)) to facilitate inter-modal interaction and capture complementary information between modalities. Importantly, the cross-Mamba module is adopted because its linear-time sequence modeling and stable state-space updates align well with the heterogeneous temporal characteristics of EEG and video data. EEG exhibits long and fine-grained temporal dynamics, whereas video features are relatively short and coarse. This state-space formulation enables robust cross-modal interaction despite disparities in temporal resolution, noise levels, and information density. The formulation can be written as:

$$\boldsymbol{g}_A, \boldsymbol{g}_B = \textit{Coss-Mam}\left(\mathbf{H}_A, \mathbf{H}_B\right) \tag{10}$$

Specifically, the cross-Mamba module utilizes a collaborative, bidirectional Mamba (Bi-Mamba) mechanism, comprising both forward and backward state space models (SSMs). The forward SSM includes $\bar{\mathbf{A}}_f$, $\bar{\mathbf{B}}_f$, $\mathbf{C}_f$, while the backward SSM contains $\bar{\mathbf{A}}_b$, $\bar{\mathbf{B}}_b$, $\mathbf{C}_b$. Among these, the bidirectional state transition matrices $\bar{\mathbf{A}}_f$ and $\bar{\mathbf{A}}_b$ are the most influential components, as they govern the temporal evolution of the latent state. In contrast, $\bar{\mathbf{B}}_f$, $\bar{\mathbf{B}}_b$, $\mathbf{C}_f$, and $\mathbf{C}_b$ are responsible for injecting input into the system and projecting hidden states to the output space.

To balance modality interaction and specificity, we share the transition matrices $\bar{\mathbf{A}}_f$ and $\bar{\mathbf{A}}_b$ across both modalities, enabling the learning of shared temporal patterns. Meanwhile, the input and output matrices ($\bar{\mathbf{B}}_f$, $\bar{\mathbf{B}}_b$, $\mathbf{C}_f$, $\mathbf{C}_b$) remain modality-specific to preserve unique modality characteristics, where $m \in \{A, B\}$. The collaborative SSM is formulated as:

$$\boldsymbol{p}_m^t = \bar{\mathbf{A}}_f\boldsymbol{p}_m^{t-1} + \bar{\mathbf{B}}_{mf}\mathbf{H}_m^t, \quad \boldsymbol{y}_m^t = \mathbf{C}_{mf}\boldsymbol{p}_m^t, \quad m \in \{A, B\} \tag{11}$$

where $\mathbf{H}$, $\boldsymbol{y}_m$, and $\boldsymbol{p}_m$ represent the input, hidden, and output features at time step t. Here, $\bar{\mathbf{A}}$ is the shared state transition matrix, while $\bar{\mathbf{B}}$ and $\mathbf{C}$ are modality-specific projection matrices. The final extracted feature $\boldsymbol{y}_m$ is passed through a linear layer followed by a fully connected layer to obtain the transformed representation. To further stabilize gradient flow and enhance feature expressiveness, we incorporate a residual pathway:

$$\boldsymbol{h}_m = LN\left(\mathcal{F}\left(\boldsymbol{p}_m\right)\right) + \mathbf{H}_m, \quad m \in \{A, B\} \tag{12}$$

$$\boldsymbol{g}_m = FFN\left(\boldsymbol{h}_m\right) + \boldsymbol{h}_m, \quad m \in \{A, B\} \tag{13}$$

By combining forward and backward processing, we obtain the fused representations $\boldsymbol{g}_A$ and $\boldsymbol{g}_B$.

Based on the above process, we obtain modality-specific features $\boldsymbol{f}_A$ and $\boldsymbol{f}_B$ and cross-modal fusion features $\boldsymbol{g}_A$ and $\boldsymbol{g}_B$. To further refine modality-level alignment, we introduce a contrastive loss that encourages fine-grained consistency and disentanglement. Specifically, we treat the pair $(\boldsymbol{f}_A, \boldsymbol{g}_A)$ (and similarly $\boldsymbol{f}_B, \boldsymbol{g}_B$) as positive pairs, as they originate from the same modality and should share semantic consistency. Therefore, their distance in the feature space should be minimized. In contrast, although $\boldsymbol{g}_A$ and $\boldsymbol{g}_B$ are obtained through cross-modal interaction, they encode complementary information from different modalities. Thus, we treat them as negative pairs and encourage their representations to be dissimilar, thereby enhancing cross-modal disentanglement:

$$\begin{aligned} \mathcal{L}_{MA} = &\left(D\left(\boldsymbol{g}_A^+, f_A\right) - D\left(\boldsymbol{g}_B^-, \boldsymbol{g}_A\right) + m, 0\right) + \\ &\left(D\left(\boldsymbol{g}_B^+, f_B\right) - D\left(\boldsymbol{g}_A^-, \boldsymbol{g}_B\right) + m, 0\right) \end{aligned} \tag{14}$$

where D represents the Jensen–Shannon distance.

3.3. Modality balancing

To address modality imbalance and prevent one modality from dominating the joint representation learning, we introduce a modality-adaptive re-initialization mechanism. Instead of relying on fixed or manually triggered resets, our method adaptively adjusts the encoder parameters based on the modality's generalization behavior, providing an automatic balancing strategy during multimodal training.

A crucial question in multimodal learning is how to determine whether a modality is well-trained, overfitted, or collapsing during training. Empirically, a well-learned representation space forms compact and well-separated clusters that remain stable across both training and unseen samples. In contrast, a collapsing or overfitted modality often produces highly concentrated clusters for the training set but exhibits noticeably degraded separability on the validation set. This observation motivates the use of the purity gap, which is defined as the difference in clustering purity between the training and validation domains, to indicate the reliability of each modality. When a modality encoder generalizes well, its feature representations naturally form clusters that correspond to emotion categories, and the purity scores on the training and validation sets are close. A large purity gap reflects poor generalization and suggests that the modality is overfitting or experiencing representation collapse. In this sense, the purity gap provides an interpretable and quantitative measure of a modality's learning status. To compute this measure, for each modality m, we split the training data into an internal training subset and a validation subset, and extract encoded representations $\mathbf{H}_m$. K-means clustering is then performed separately on both subsets, and the number of clusters K is set to the number of emotion categories. For the training set, the encoded samples are denoted as $\mathcal{M}\left(\mathbf{H}_m\right) = \left\{\varphi_1^m, \varphi_2^m, \ldots, \varphi_{N_D}^m\right\}$, and the corresponding cluster centroids are $\mathbf{C} = \left\{C_1, C_2, \ldots, C_K\right\}$.

In the assignment step, each sample is assigned to the cluster whose centroid is closest in Euclidean distance. Specifically, a given sample φ_i^m is assigned to the cluster C_k with centroid O_k if:

$$\left\|\varphi_i^m - O_k\right\|^2 \le \left\|\varphi_i^m - O_j\right\|^2, \forall j \in \{1, 2, \ldots, K\} \tag{15}$$

where $\|\cdot\|$ denotes the L_2-norm. In the update step, each cluster centroid is recalculated as the mean of the features assigned to it:

$$O_k = \frac{1}{|C_k|}\sum_{\varphi_i^m \in C_k}\varphi_i^m \tag{16}$$

This assignment and update process iterates until convergence (i.e., the assignments stabilize or a predefined maximum number of iterations is reached), yielding the final clustering results.

To quantify the reliability of the learned representation of the modality m, we adopt clustering purity, a widely used metric to evaluate clustering performance. The encoded features $\mathbf{H}_m$ are grouped according to their ground truth labels into K classes, denoted as $\mathbf{P} = \{P_1, P_2, \ldots, P_K\}$. The training set purity is computed as:

$$V_D^k = \frac{1}{N_D} \sum_{C_k \in \mathbf{C}} \max_{P_m \in \mathbf{P}} |C_k \cap P_k| \tag{17}$$

Similarly, the clustering purity on the validation set is denoted as V_T^k. We then define the generalization gap for modality m as the absolute difference between training and validation purity. To avoid abrupt parameter resets that may harm convergence, we regulate the strength of re-initialization with a smooth and bounded mapping:

$$l_k = \left| V_D^k - V_T^k \right| \tag{18}$$

$$\alpha_m = tanh(\lambda \cdot l^m) \tag{19}$$

where $\alpha_m \in [0, 1)$ controls the strength of re-initialization, and λ adjusts the sensitivity of the mapping. The *tanh* function ensures that re-initialization strength increases gradually and saturates for large gaps, preventing oscillations or disruptive updates. This design ensures stability and avoids oscillations or frequent disruptive updates. In practice, we set $\lambda = 3$, which provided consistent behavior across all datasets.

Finally, instead of fully resetting the encoder, we update its parameters through a convex combination of the current parameters and the initial parameters:

$$\theta_m = (1 - \alpha_m) \cdot \theta_m^{cur} + \alpha_m \cdot \theta_m^{int} \tag{20}$$

where θ_m^{cur} and θ_m^{int} denote the current and original initialized parameters of the encoder, respectively. This mechanism ensures that modality-adaptive balancing is stable, incremental, and preserves accumulated knowledge. Each modality autonomously regulates its update strength according to its learning dynamics, making re-initialization a continuous, self-correcting process rather than an externally triggered discrete event.

3.4. Joint optimization

Finally, we integrate the self-attention features and cross-modal fusion features from both modalities for classification. The fused feature $\mathbf{Z}$ is then fed into a classification module, which is optimized using the standard cross-entropy loss:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{N} \mathbf{Y}_i log\left(\mathbf{Y}_i^{'}\right). \tag{21}$$

where $\mathbf{Y}_i^{'}$ is the predicted probability vector for the ith sample, and $\mathbf{Y}_i$ is the corresponding one-hot ground truth label.

The overall training objective incorporates the classification loss, the distribution-level alignment loss $\mathcal{L}_{DA}$, and the modality-level alignment loss $\mathcal{L}_{MA}$:

$$\mathcal{L} = \mathcal{L}_{CE} + \gamma_1 \mathcal{L}_{DA} + \gamma_2 \mathcal{L}_{MA} \tag{22}$$

where γ_1 and γ_2 are the hyper-parameters. The specific process is as shown in Algorithm 1.

Algorithm 1: The training process of MAAB-ER.

Input: Shallow feature of modality $\mathbf{X}_m \in \mathbb{R}^{N \times L_m}$, epoch $\mathcal{E}$, learning rate l.
Output: Classification Results $\mathcal{R}$

Forward Noising Training;
Initialize model parameters θ;
for *iteration* $= 1, 2, \ldots, \mathcal{E}$ **do**
  Acquire the modality-specific features via Eq. (1);
  Acquire the self-attention and cross fusion features via Eq. (9) and Eq. (10);
  Calculate the alignment loss via Eq. (7) and Eq. (14);
  Acquire the total loss via Eq. (22);
  Update the model parameter $\theta \leftarrow \theta - l \times \nabla \mathcal{L}$;
  Re-initizliza the model-specific parameters via Eq. (20);
return Classification results $\mathcal{R}$.

4. Experiment results and analysis

4.1. Datasets

4.1.1. DEAP

The DEAP dataset [34] records EEG signals and facial expressions from 32 participants while they watched 40 one-minute-long music video clips. The EEG signals were captured using 32 channels. However, due to experimental constraints, facial expression data were only recorded for 22 participants. To ensure modality alignment in our experiments, we selected the EEG recordings corresponding to the participants with available facial expression data. The original DEAP dataset provides preprocessed EEG signals, including downsampling, artifact removal, and band-pass filtering. Based on these signals, we further decomposed the EEG into four standard frequency bands: *theta* (4.0–8.0 Hz), *alpha* (8.0–13.0 Hz), *beta* (13.0–30.0 Hz), and *gamma* (30.0–45.0 Hz).

4.1.2. SEED-IV

The SEED-IV dataset [35] contains 62-channel EEG and eye movement data from 15 participants who watched 72 emotional video clips. The experiment was divided into three sessions, each consisting of 24 videos, with at least three days between sessions to avoid emotional carryover effects. In this study, we utilized the preprocessed data provided by the dataset. The preprocessing includes downsampling, band-pass filtering, and artifact reduction. For EEG signals, we used five standard frequency bands: *delta* (1.0–4.0 Hz), *theta* (4.0–8.0 Hz), *alpha* (8.0–14.0 Hz), *beta* (14.0–31.0 Hz), and *gamma* (31.0–50.0 Hz). For eye movement data, we used 30 statistical features extracted from the preprocessed recordings.

4.1.3. SEED-V

The SEED-V dataset [36] is an extension of the SEED series for visual-induced emotion recognition. It contains 62-channel EEG recordings from 16 participants while watching 15 video clips, each associated with one of five discrete emotions. We used the publicly available preprocessed EEG signals, where noise-contaminated segments have been mitigated through standard preprocessing procedures such as filtering and downsampling. The EEG signals were decomposed into five frequency bands: *delta* (1.0–4.0 Hz), *theta* (4.0–8.0 Hz), *alpha* (8.0–14.0 Hz), *beta* (14.0–31.0 Hz), and *gamma* (31.0–50.0 Hz). For eye movement data, we directly utilized the provided 30-dimensional statistical feature vectors.

4.2. Experimental setting

To ensure the robustness and generalizability of the proposed method, all experiments are conducted using five-fold cross-validation. For each dataset, subjects are randomly partitioned into five folds. In each iteration, one fold is used as the test set, while the remaining four folds are used for training. Within the training folds, a subset is further split as a validation set for hyperparameter tuning and model selection. The test set is strictly held out for final evaluation and is not involved in any training or model adjustment process. To address modality imbalance, we perform adaptive re-initialization of modality-specific encoder parameters every 20 training epochs, with a maximum of 10 re-

Table 2
Comparison with alignment and balance multimodal learning methods on the DEAP, SEED-IV, and SEED-V datasets.

Category	Methods	Description	Datasets
Alignment	CMSLNet [37]	Proposes an instance-level adaptive consistency metric to align EEG and eye-movement signals.	SEED-IV/V
	AEKM [38]	Employs empirical kernel mapping with SVM to fuse heterogeneous multimodal features.	SEED-IV/V
	DCCA [36]	Learns correlated latent representations via deep canonical correlation analysis.	DEAP/SEED-IV/V
	Subnetwork [39]	Constructs label-wise averaged brain networks to suppress weak neural associations.	DEAP/SEED-IV
	Cross [40]	Introduces a cross-modal guidance network to capture complementary information.	SEED-V
	DG-JCA [16]	Integrates topology, consistency, and complementarity using joint cross-attention.	DEAP/SEED-IV
	MMV-SSTMA-Att [41]	Applies self-supervised spatiotemporal attention for EEG-eye-movement fusion.	SEED-IV/V
Balance	CKD [9]	Utilizes counterfactual causal estimation to upweight weaker modalities.	DEAP/SEED-IV/V
	PMR [11]	Enhances slow-learning modalities via prototypical rebalancing.	DEAP/SEED-IV/V
	AGM [42]	Modulates gradients adaptively to stabilize multimodal optimization.	DEAP/SEED-IV/V
	DLMG [43]	Integrates contrastive and supervised objectives to improve modality balance.	DEAP/SEED-IV/V
	OGM-GE [8]	Conducts on-the-fly gradient modulation by monitoring modality discrepancy.	DEAP/SEED-IV/V
	MTAMW [44]	Introduces a multimodal adaptive weight matrix that dynamically allocates appropriate weights according to modality contribution.	DEAP/SEED-IV/V
	OPM [45]	Suppresses dominant modalities by stochastically dropping their features with dynamic probability in the feed-forward stage.	DEAP/SEED-IV/V
	OGM [45]	Mitigates dominance by attenuating the gradient of stronger modalities during back-propagation.	DEAP/SEED-IV/V
	Joint	Direct concatenation of modality-specific features for unified classification.	DEAP/SEED-IV/V

estimations. The re-initialization is triggered based on the purity gap between the training and validation sets, and the update strength is regulated by a smooth bounded mapping controlled by the hyperparameter λ. The alignment strength is regulated by two hyperparameters, γ_1 and γ_2, which are empirically set according to the dataset characteristics: $\gamma_1 = 2$ and $\gamma_2 = 0.5$ for DEAP, $\gamma_1 = 3$ and $\gamma_2 = 3$ for SEED-IV, and $\gamma_1 = 3$ and $\gamma_2 = 1$ for SEED-V. Detailed analysis and discussion regarding the influence of these hyperparameters are provided in the Discussion section.

All experiments are implemented in PyTorch 2.0.1 and conducted on a workstation equipped with an NVIDIA RTX 4080 GPU. The Adam optimizer is employed with an initial learning rate of 1×10^{-4} and a batch size of 64. Early stopping is applied based on the validation loss to prevent overfitting.

4.3. Comparison with state-of-the-art methods

To comprehensively evaluate the effectiveness of the proposed alignment and balance mechanisms, we selected several representative multimodal alignment and fusion methods from recent years as comparison baselines. These methods cover both feature alignment and modality balancing, providing a thorough reference for assessing the advantages of our approach on key tasks. A summary of all comparison methods is provided in Table 2.

4.3.1. The comparison with alignment methods

To verify the effectiveness of the proposed modality alignment mechanism, we compare our method with several representative MER approaches from recent literature. To ensure a fair comparison, all methods are evaluated on the same datasets under identical experimental settings. The results are summarized in Table 3.

As illustrated in the table, our proposed MAAB-ER model outperforms all existing alignment-based methods across most dimensions and datasets. The only exceptions occur in the valence dimension and the arousal-valence dimension of the DEAP dataset, where DCCA [36] and DG-JCA report slightly higher scores by 2.02% and 2.56%, respectively. Overall, these results demonstrate that our alignment strategy and adaptive re-initialization mechanism substantially enhance multimodal fusion and classification accuracy. The marginal performance gap on DEAP-V and DEAP-AV is reasonable given the inherent challenges in MER. Emotional responses exhibit strong subjectivity and high intra-class variability, and different modalities often contribute unevenly across emotional dimensions. We speculate that the advantage of DCCA may stem from its canonical-correlation-based representation learning, which is particularly effective for modeling stable cross-modal relationships in valence-related tasks. Meanwhile, DG-JCA likely enforces instance-level structural consistency across modalities, providing an advantage on tasks with more stable emotional patterns. In contrast, MAAB-ER focuses on bridging both distributional and semantic gaps through global alignment, while simultaneously rebalancing modality contributions through adaptive gradient modulation. This enables the model to achieve broader generalization across more diverse and complex emotional dimensions. In summary, the results verify the effectiveness and robustness of the proposed MAAB-ER framework. By integrating cross-modal alignment with adaptive modality balancing, our model consistently achieves state-of-the-art performance and effectively addresses the problems of distribution discrepancies and modality imbalance.

4.3.2. The comparison with balance methods

To verify the effectiveness of our proposed method in addressing modality imbalance, we compare it with several recently developed

Table 3
Comparison with alignment multimodal learning methods on the DEAP, SEED-IV, and SEED-V datasets.

Methods	DEAP-A	DEAP-V	DEAP-AV	SEED-IV	SEED-V
CMSLNet [37]	–	–	–	83.15	87.32
AEKM [38]	–	–	–	79.41	79.84
DCCA [36]	84.30	**95.60**	–	87.50	85.30
Subnetwork [39]	85.34	86.61	–	–	84.51
Cross [40]	–	–	–	–	90.21
DG-JCA [16]	92.89	92.50	**91.65**	88.24	–
MMV-SSTMA-Att [41]	–	–	–	90.16	81.38
MAAB-ER	**94.43**	93.58	89.09	**91.48**	**90.56**

imbalance-handling strategies, including CKD [9], PMR [11], AGM [42], DLMG [43], OGM-GE [8], MTAMW [44], OPM [45], OGM [45], as well as a classical joint-optimization approach ("Joint"), which concatenates modality-specific features for unified classification and serves as a widely adopted baseline for evaluating modality imbalance. From the results presented in Table 4, we observe that all these imbalance-aware methods achieve competitive performance, confirming that modality imbalance is indeed a critical and practical issue in MER. This highlights the necessity of explicitly addressing this challenge. Notably, our proposed MAAB-ER consistently achieves the highest classification accuracy across all datasets. Furthermore, it obtains the lowest standard deviation in most cases, indicating its superior robustness and generalization capability. These results validate the importance of jointly tackling both modality alignment and imbalance. Our modality-adaptive re-initialization mechanism plays a key role in this success by dynamically adjusting the parameters of modality-specific encoders based on their learning effectiveness. This strategy enables the model to balance modality contributions more adaptively, avoiding excessive reliance on noisy or uninformative modalities

Additionally, existing methods commonly follow a heuristic of amplifying weaker modalities while suppressing stronger ones, which can be effective in balanced or low-complexity settings. However, as the complexity of the tasks increases, their performance will significantly deteriorate and usually fail to reach the performance level of MAAB-ER. This suggests that simple weight reallocation strategies are insufficient for handling the intricate cross-modal dependencies and semantic divergences present in more complex scenarios. Importantly, these competing methods often neglect the issue of modality misalignment. Without explicitly aligning the feature distributions and semantic spaces of different modalities, their fusion strategies risk introducing noise and redundancy, which undermines overall performance. In contrast, MAAB-ER incorporates a dual-level alignment module that performs both distribution-level and modality-level alignment. This helps bridge structural and semantic gaps across modalities. Coupled with our modality-adaptive re-initialization strategy, MAAB-ER achieves robust and balanced optimization by reinforcing underperforming modalities and preventing overfitting in stronger ones. Together, these mechanisms allow the model to consistently outperform existing approaches across diverse datasets and classification challenges.

4.4. Ablation study

4.4.1. The contribution of alignment and re-initialization

The proposed model comprises two core components: the alignment module and the re-initialization module. To assess the effectiveness of each component, we perform comprehensive ablation studies. Specifically, we evaluate the model's performance after individually removing each module. For the alignment module, we also investigate the individual contributions of the alignment loss terms $\mathcal{L}_{MA}$ and $\mathcal{L}_{DA}$ by selectively excluding them. Additionally, we conduct paired t-tests to assess the statistical significance of the performance differences. The results are summarized in Table 5.

As illustrated in the table, removing both $\mathcal{L}_{MA}$ and $\mathcal{L}_{DA}$, which effectively eliminates the entire alignment module, leads to a substantial performance drop compared with configurations that use only $\mathcal{L}_{CE} + \mathcal{L}_{DA}$ or $\mathcal{L}_{CE} + \mathcal{L}_{MA}$. The decline is even more pronounced when compared with the full model that incorporates all three loss terms. These results demonstrate that the alignment losses play a critical role in reducing both distributional and semantic discrepancies between modalities, thereby improving the coherence of shared representations. Likewise, removing the re-initialization module also causes a clear performance decrease, which confirms its ability to adaptively alleviate modality imbalance and enhance generalization. In addition, the *t*-test results show that most components, including the individual alignment losses and the re-initialization module, exhibit statistically significant differences relative to the complete model. This finding highlights the unique and indispensable contribution of each component and further supports the necessity of integrating both modules within the proposed framework.

Notably, incorporating only the alignment module or only the re-initialization module does not yield consistent performance improvements across all datasets. In contrast, combining both modules leads to significant and consistent gains. This observation suggests that the two modules are complementary: the alignment module improves cross-modal feature consistency, while the re-initialization module adaptively redistributes learning capacity to underrepresented modalities. Together, they form a synergistic mechanism that substantially enhances the robustness and overall performance of the model.

Moreover, we analyze the ROC curves of the model under different configurations, as shown in Fig. 3. The results reveal that for the DEAP dataset, removing the alignment module causes more severe performance degradation than removing the re-initialization module. Conversely, for the SEED-IV and SEED-V datasets, the re-initialization module has a greater impact, with its removal leading to larger performance drops than the removal of the alignment module. These trends further confirm that the two modules address distinct modality-specific challenges and complement each other. Collectively, these findings validate the effectiveness of our dual-module design, which integrates alignment and re-initialization to achieve robust and adaptable multimodal learning.

4.4.2. The contribution of alignment loss

We further conduct ablation experiments on each individual alignment loss within the alignment module. Specifically, we remove the re-initialization module so that the model includes only the classification loss and alignment-related losses. This allows us to independently evaluate the contribution of each alignment term. The results are summarized in Table 6.

As shown in the table, adding the distribution-level loss $\mathcal{L}_{DA}$ on top of the baseline classification loss $\mathcal{L}_{CE}$ leads to a more significant performance improvement than adding the modality-level loss $\mathcal{L}_{MA}$ alone. This indicates that adjusting the feature distribution of the original data is more important than enforcing semantic consistency between modality-specific and cross-modal features. In other words, distribution-level alignment offers a stronger regularization effect, thereby enhancing the generalization capability of the learned representations.

Moreover, the best performance is generally achieved when all three loss functions $\mathcal{L}_{CE}$, $\mathcal{L}_{DA}$, and $\mathcal{L}_{MA}$ are used together, indicating the complementary nature of distribution- and modal-level alignment. However, an exception is observed on the SEED-V dataset, where the combination of $\mathcal{L}_{CE}$ and $\mathcal{L}_{DA}$ yields the best result, while introducing $\mathcal{L}_{MA}$ slightly degrades performance. We hypothesize that this may be due to the inherently noisy or weakly correlated relationship between modality-specific and cross-modal features in SEED-V, which makes modality-level alignment less effective or even harmful.

In addition, we conduct paired t-tests to examine the statistical significance of performance differences among the configurations. The results align with the aforementioned observations, further confirming that the performance gains from $\mathcal{L}_{DA}$ are statistically more significant

Table 4
Comparison with imbalanced multimodal learning methods (accuracies ± standard deviation) on the DEAP, SEED-IV, and SEED-V datasets.

Methods	DEAP-A	DEAP-V	DEAP-AV	SEED-IV	SEED-V
CKD [9]	76.65 ± 21.23	80.11 ± 16.32	72.16 ± 16.62	82.87 ± 8.23	89.31 ± 2.82
PMR [11]	93.24 ± 2.27	92.73 ± 2.26	70.51 ± 20.51	87.92 ± 2.62	88.54 ± 3.96
AGM [42]	93.35 ± 2.57	79.55 ± 17.89	87.05 ± 1.71	86.02 ± 5.32	89.24 ± 2.84
DLMG [43]	85.23 ± 16.95	86.36 ± 14.08	86.70 ± 1.85	87.59 ± 3.99	88.96 ± 2.71
OGM-GE [8]	93.07 ± 1.20	79.43 ± 17.77	86.82 ± 2.19	86.25 ± 5.81	89.31 ± 2.83
MTAMW [44]	90.78 ± 6.48	91.57 ± 2.57	89.47 ± 4.81	88.47 ± 6.18	86.48 ± 7.81
OPM [45]	88.48 ± 4.51	87.58 ± 4.18	84.71 ± 6.76	85.76 ± 4.89	83.48 ± 6.87
OGM [45]	89.72 ± 4.18	88.81 ± 4.95	87.84 ± 86.72	88.84 ± 6.75	89.72 ± 6.57
Joint	92.67 ± 2.93	92.89 ± 3.11	86.02 ± 2.32	90.57 ± 1.58	88.13 ± 1.35
MAAB-ER	**94.43 ± 1.58**	**93.58 ± 1.94**	**89.09 ± 2.33**	**91.48 ± 2.07**	**90.56 ± 0.16**

The results of the above self-replication.

Table 5
The ablation results on DEAP, SEED-IV, and SEED-V datasets with statistical significance (* denotes $p < 0.05$ from t-test).

Methods	DEAP-A	DEAP-V	DEAP-AV	SEED-IV	SEED-V
$\mathcal{L}_{CE}$	85.46* ± 17.18	75.39* ± 18.26	80.32* ± 9.12	88.86* ± 2.00	87.80* ± 5.76
$\mathcal{L}_{CE} + \mathcal{L}_{DA}$	93.58 ± 1.56	92.22 ±1.89	86.53* ± 1.84	90.69* ± 1.22	88.61* ± 2.74
$\mathcal{L}_{CE} + \mathcal{L}_{MA}$	85.51* ± 15.26	76.71* ± 17.91	81.25* ± 14.69	88.84* ± 0.72	78.13* ± 5.86
w/o Re-initialization	91.71* ± 2.55	91.25* ± 1.89	86.65* ± 2.34	89.72* ± 2.23	86.88* ± 1.57
MAAB-ER	**94.43 ± 1.58**	**93.58 ± 1.94**	**89.09 ± 2.33**	**91.48 ± 2.07**	**90.56 ± 0.16**

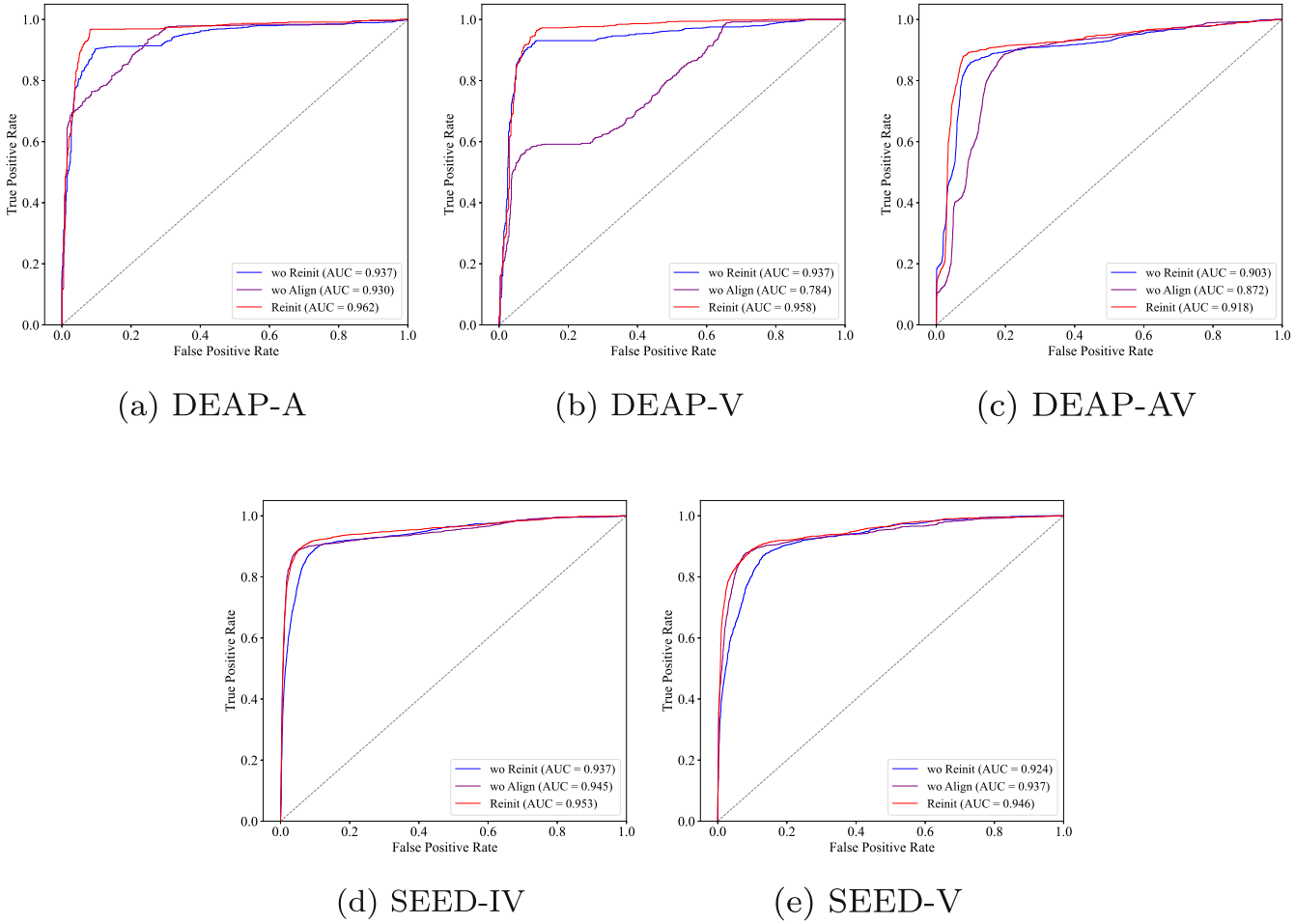


Fig. 3. The ablation ROC curves of the alignment module and the re-initialization module on the DEAP, SEED-IV and SEED-V datasets.

Table 6
The ablation results of the alignment module (without re-reinitialization) on DEAP, SEED-IV, and SEED-V datasets with statistical significance (* denotes $p < 0.05$ from t-test).

Methods	DEAP-A	DEAP-V	DEAP-AV	SEED-IV	SEED-V
$\mathcal{L}_{CE}$	77.05* ± 20.20	82.50* ± 15.03	70.17* ± 14.23	87.42* ± 15.54	85.31* ± 9.76
$\mathcal{L}_{CE} + \mathcal{L}_{DA}$	89.51* ± 3.00	84.94* ± 14.74	86.19* ± 1.91	89.19 ±1.32	**89.93*** ± 2.29
$\mathcal{L}_{CE} + \mathcal{L}_{MA}$	80.23* ± 15.95	84.72* ± 13.85	80.97* ± 15.64	89.31 ± 10.26	84.93 ± 8.24
$\mathcal{L}_{CE} + \mathcal{L}_{DA} + \mathcal{L}_{MA}$	**91.71 ± 3.40**	**91.25 ± 2.60**	**86.65** ± 1.49	**89.72** ± 1.69	86.88 ± 4.64

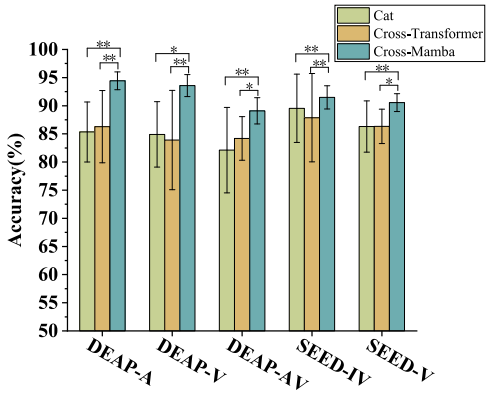


Fig. 4. The performance of fusion methods on the DEAP, SEED-IV, and SEED-V datasets. "*" and "**" indicate statistical significance at the $p < 0.05$ and $p < 0.01$ levels, respectively, based on paired t-tests.

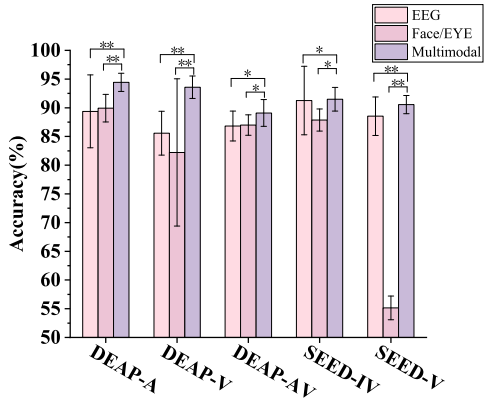


Fig. 5. The performance of different modalities on the DEAP, SEED-IV, and SEED-V datasets. "*" and "*&" indicate statistical significance at the $p < 0.05$ and $p < 0.01$ levels, respectively, based on paired t-tests.

than those from $\mathcal{L}_{MA}$ in most cases, and that combining both alignment losses yields superior performance with statistical confidence.

Despite this exception, the overall trend across the majority of datasets supports the effectiveness of our alignment strategy. These results demonstrate that both alignment losses contribute to performance improvement, and their combination generally offers the most robust solution, validating the essential role of the alignment module in our framework.

4.4.3. The contribution of cross-Mamba

To evaluate the effectiveness of the proposed cross-Mamba module, we conduct ablation studies in which it is removed and replaced by two alternative designs: (1) a simple feature concatenation (Cat), and (2) a standard cross-modal Transformer. The corresponding results are presented in Fig. 4, demonstrating the comparative performance of these variants.

From the results shown in Fig. 4, both the simple concatenation (Cat) baseline and the cross-modal Transformer yield consistently lower performance than the proposed Cross-Mamba module across all datasets. Moreover, the differences are statistically significant according to paired t-tests. Compared with Cat, cross-Mamba achieves substantially higher accuracy and stability because Cat performs no explicit interaction modeling and thus fails to capture complementary information between modalities. In contrast, cross-Mamba leverages bidirectional state-space interactions to model cross-modal dependencies in a structured and temporally aware manner. Compared with the cross-modal Transformer, cross-Mamba also demonstrates clear advantages. Transformers rely on quadratic-complexity attention and tend to struggle with long EEG sequences, which may lead to noisy or unstable alignment when paired with shorter video features. The state-space formulation of Cross-Mamba enables linear-time sequence modeling, smoother state transitions, and better robustness to differences in temporal resolution, noise level, and information density between modalities. This leads to more reliable and better-aligned cross-modal representations.

4.5. Discussion

4.5.1. Unimodal performance evaluation

We further conduct ablation experiments on each modality to investigate the contribution of different unimodal signals to the overall performance. As illustrated in Fig. 5, the unimodal models for each dataset consistently perform worse than the multimodal model, confirming the effectiveness of cross-modal integration.

In most cases, EEG-based models outperform those relying solely on eye movements or facial expressions, suggesting that EEG signals carry more discriminative emotional information. However, even the best unimodal performance remains significantly lower than that of the full multimodal model. This result highlights the complementary nature of different modalities and supports the conclusion that combining EEG with eye or facial features enables the model to better capture the complexity of emotional states, leading to superior classification performance.

4.5.2. Comparison in weak-modality scenarios

To further evaluate the effectiveness of the proposed method in addressing modality imbalance, we conduct a robustness analysis by injecting controlled noise into individual modalities, simulating extreme modality degradation scenarios. Specifically, Gaussian white noise with zero mean is added to randomly selected samples, where 80% of the data in the target modality are corrupted. The noise intensity is controlled by setting the standard deviation to 0.8 relative to the normalized signal range. This setting corresponds to a high-intensity noise condition, where the majority of samples are contaminated by high-amplitude noise, resulting in severe degradation of modality quality. This quantitative definition ensures that the noise level is clearly specified and reproducible. This setup allows us to examine whether the proposed modality-adaptive re-initialization mechanism can still maintain stable and balanced learning under severe modality imbalance. Additionally, we compare our method with several state-of-the-art imbalance handling approaches, including CKD [9], PMR [11], AGM [42], DLMG [43], OGM-GE [8], and Joint. The results are illustrated in Fig. 6.

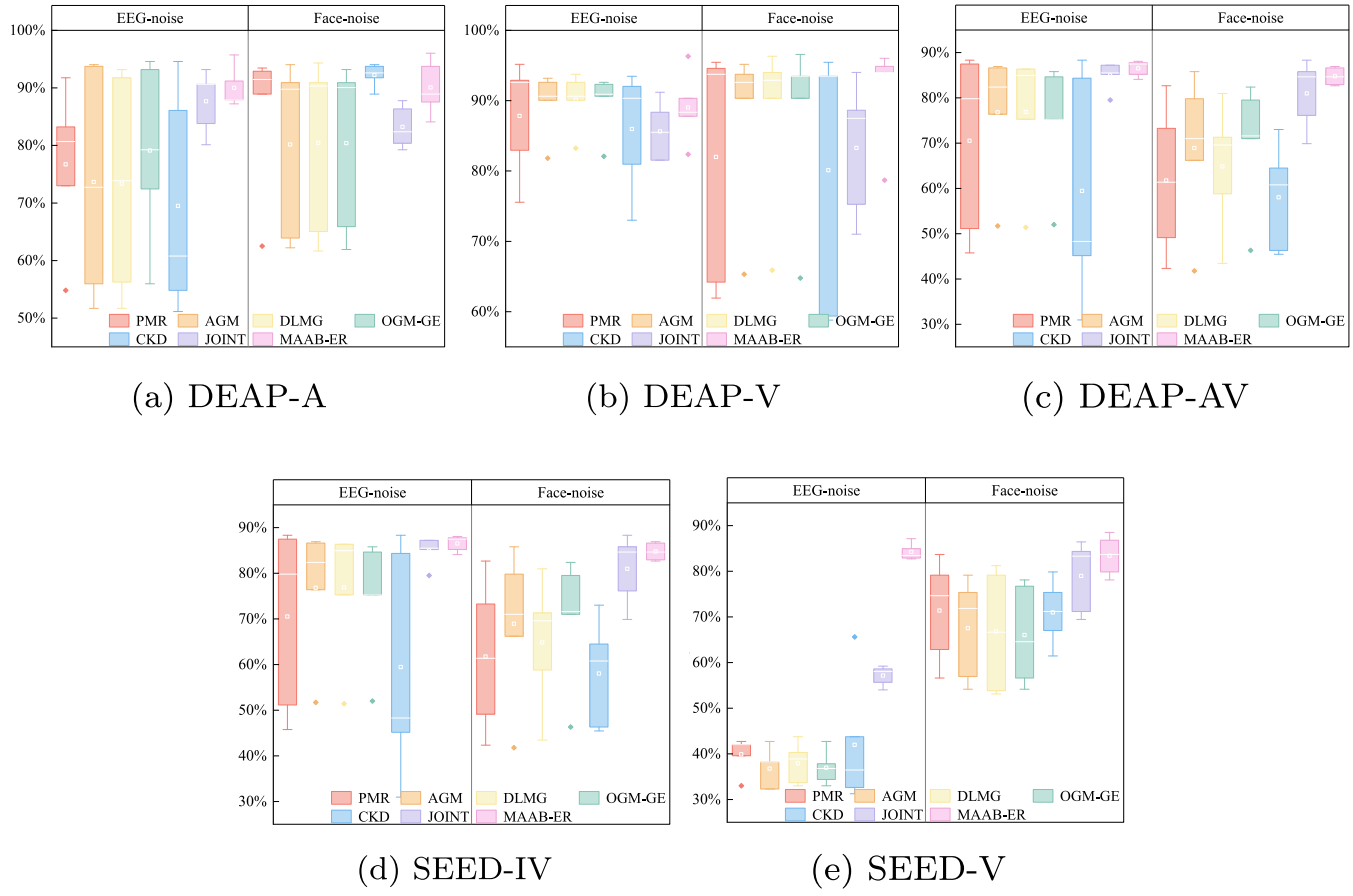


Fig. 6. Classification accuracies using imbalanced methods under scarcely informative modality conditions on the DEAP, SEED-IV, and SEED-V datasets.

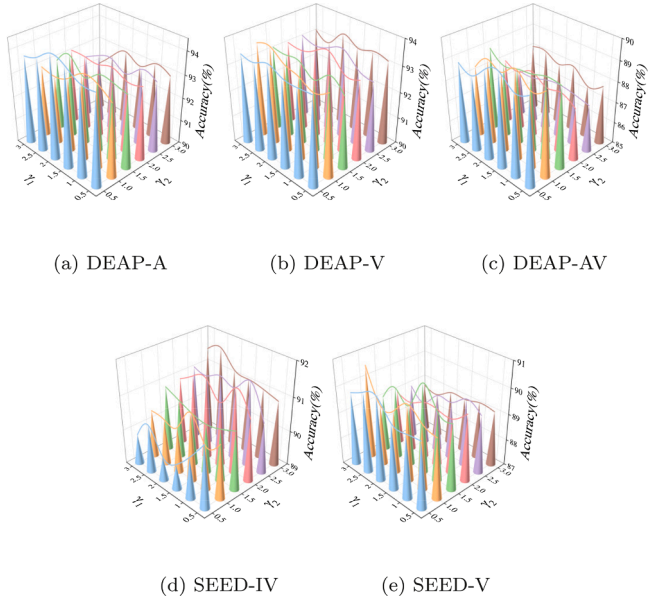


Fig. 7. The results of hyper-parameters γ_1 and γ_2 on the DEAP, SEED-IV, and SEED-V datasets.

From the figure, we observe that when a modality is heavily corrupted by noise, the overall data distribution is significantly distorted, leading to pronounced modality imbalance. As a result, existing imbalance-handling methods experience substantial drops in classification accuracy, particularly in more complex tasks such as multi-class classification. This highlights the limitations of these methods in adapting to extreme inter-modal inconsistencies. In contrast, our proposed MAAB-ER model demonstrates strong resilience. While its performance does decline slightly under severe noise conditions, the degradation is relatively small, and it consistently maintains competitive performance across all datasets and task complexities. This suggests that MAAB-ER effectively mitigates the negative effects of modality imbalance, even in the presence of substantial noise.

Notably, there are a few edge cases where MAAB-ER performs poorly compared to specific baselines. For example, when facial features are corrupted, the performance on the arousal dimension is marginally lower than that of CKD. Similarly, when EEG signals are replaced with noise, the performance on the valence dimension of the DEAP dataset is slightly lower than that of DLMG. We hypothesize that these methods may have specialized optimization mechanisms favoring certain modalities, leading to better results in specific scenarios. However, such advantages tend to be task-dependent and lack generalizability. Overall, despite these isolated instances, MAAB-ER achieves the highest overall accuracy and lowest performance variance in most conditions. These findings validate the robustness and general applicability of our proposed dual-strategy framework, which jointly addresses modality alignment and adaptive imbalance correction, even under extreme modality degradation.

4.5.3. *Computational efficiency and convergence analysis*

To further evaluate the practicality of the proposed MAAB-ER framework, we analyze its computational overhead, parameter scale, and convergence behavior in comparison with the baseline multimodal model. Although our approach introduces two additional components, namely the alignment module and the re-initialization mechanism, the overall computational burden remains lightweight.

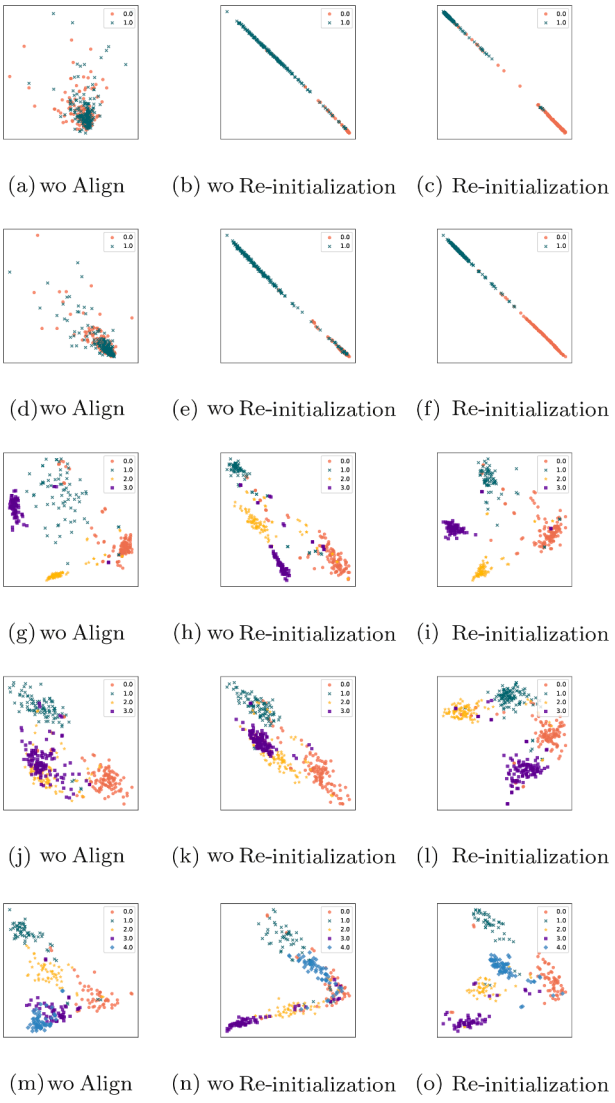


Fig. 8. t-SNE Visualization of Feature Distributions under Three Settings: Without Alignment, Without Re-initialization, and Full Model (Rows 1–5 correspond to DEAP-A, DEAP-V, DEAP-AV, SEED-IV, and SEED-V).

Table 7
Comparison of computational cost. (measured on an RTX 4080 GPU).

Method	Params (M)	Epoch time (s)	Train time (min)
Baseline	68.484	2.4	13.2
Alignment	69.147	2.8	14.7
Re-initialization	68.998	2.5	13.5
MAAB-ER	**69.935**	**3**	**16**

Table 7 summarizes the number of trainable parameters and the average training cost measured on an RTX 4080 GPU. As shown, the alignment module slightly increases the parameter count (approximately 1.0M), while the re-initialization mechanism does not introduce additional learnable parameters. The full MAAB-ER model results in a parameter increase of only 1.4% relative to the baseline, indicating that the architecture remains highly compact. In terms of computational cost, the average time per epoch increases only marginally (from 2.4 s to 3.0 s), and the total training time rises from 13.2 min to 16 min. This overhead is mainly attributed to the forward pass of the alignment module and the occasional re-initialization operation, both of which involve lightweight matrix transformations rather than computationally expensive attention mechanisms. Therefore, the added cost is acceptable and has minimal impact on model scalability. Additionally, we observe consistent convergence behavior across all folds, where the training and test losses decrease smoothly without oscillation, and the proposed framework exhibits more stable convergence than the baseline. The alignment module facilitates more consistent multimodal representation learning, while the re-initialization mechanism mitigates early-stage overfitting by preventing suboptimal feature dominance. Consequently, the MAAB-ER model reaches lower test loss and higher final accuracy with only a minor increase in computational cost.

Overall, these results demonstrate that the proposed MAAB-ER framework achieves substantially improved multimodal alignment and classification performance with only a marginal increase in computational cost. The model remains lightweight, efficient, and stable during optimization, confirming its practicality for real-world multimodal affective computing applications.

4.5.4. Hyperparameter sensitivity

We perform a grid search to tune the hyperparameters of the alignment loss function for each dataset. The results for different emotional dimensions across datasets are presented in Fig. 7. As illustrated in the figure, the performance variation under different hyperparameter combinations is generally within a 2% margin, indicating that the proposed alignment strategy is stable and not overly sensitive to the choice of hyperparameters.

For the DEAP dataset, the optimal results for arousal, valence, and the joint arousal-valence dimension are consistently obtained with $\gamma_1 = 2$ and $\gamma_2 = 0.5$. This suggests that moderate distribution-level alignment, combined with a weaker semantic-level constraint, provides the best cross-modal representation in this setting. For the SEED-IV dataset, the optimal configuration is $\gamma_1 = 3$ and $\gamma_2 = 3$, implying that stronger distribution and semantic alignment are beneficial due to the larger modality discrepancies present in the data. In contrast, for the SEED-V dataset, the best performance is achieved with $\gamma_1 = 3$ and $\gamma_2 = 1$, indicating that strong distribution alignment remains important, while a relatively relaxed semantic alignment constraint is more suitable. This is likely because the semantic correlation between modalities in SEED-V is weaker, and imposing a strict semantic alignment may introduce additional noise.

Overall, these results confirm the robustness and effectiveness of our alignment mechanism across different multimodal scenarios. They also highlight the necessity of flexibly adjusting γ_1 and γ_2 to adapt to dataset-specific fusion characteristics and semantic relationships.

4.6. Visualization analysis

To further validate the effectiveness of the proposed alignment and re-initialization modules, we conduct t-SNE visualizations of the learned feature distributions under three different settings: without alignment, without re-initialization, and the complete model. The results are shown in Fig. 8, where each row corresponds to a specific dataset, allowing direct comparison across different configurations.

From the figure, we observe that when the alignment module is removed, the feature distributions exhibit substantial class overlap and weak class separability. This issue is especially evident on the DEAP dataset, where the boundaries between categories become noticeably blurred. This suggests that the alignment module plays a critical role in promoting discriminative and semantically coherent representations. Similarly, when the re-initialization module is removed, the feature distributions also suffer from increased class overlap or less distinct clustering. This phenomenon is especially evident on the SEED-IV and SEED-V datasets, indicating that the re-initialization mechanism is essential for balancing modality-specific learning and enhancing generalization.

In summary, these visual results support our claim that both the alignment and re-initialization modules are indispensable components.

They play complementary roles in improving the quality and separability of the learned feature space across different datasets.

5. Conclusion

This paper proposes an MER model that jointly addresses modality alignment and modality balancing, effectively tackling the challenges of feature distribution gaps and learning imbalance across modalities. Unlike conventional approaches that tend to enhance weak modalities while suppressing strong ones, our method introduces a modality-adaptive re-initialization strategy that adaptively updates the parameters of modality-specific encoders based on each modality's learning capacity. This enables true dynamic balancing and mitigates issues such as overfitting and redundancy. Extensive experiments are conducted on DEAP, SEED-IV, and SEED-V datasets, achieving classification accuracies of 94.43%, 91.48%, and 90.56%, respectively. Furthermore, ablation studies verify the individual contributions of each component, showing that both the alignment and re-initialization modules significantly improve performance, with clear statistical differences. Additionally, t-SNE visualizations reveal that the alignment module effectively narrows inter-modal distributional gaps, while the re-initialization mechanism enhances the discriminability of fused representations. Together, these findings validate the effectiveness and robustness of the proposed framework in MER.

Although the proposed MAAB-ER framework demonstrates strong performance in addressing modality alignment and imbalance in MER, it currently assumes fixed modality combinations, such as EEG and eye movement. In the future, the framework will be extended to flexibly handle missing modalities or dynamically changing modality sets, thereby enhancing its applicability in real-world scenarios. Additionally, in future work, we plan to investigate cross-dataset generalization, for example, training on SEED-IV and testing on DEAP, as a way to evaluate the robustness and adaptability of our alignment and balancing mechanisms under domain shifts.

CRediT authorship contribution statement

Cheng Cheng: Writing – original draft, Methodology, Conceptualization; **Wenzhe Liu:** Visualization, Investigation; **Ruisi Shang:** Validation, Investigation; **Yikang Cheng:** Visualization, Formal analysis; **Ziyu Jia:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the anonymous reviewers for their insightful comments and suggestions to improve the quality of this paper significantly. This work was supported by the National Natural Science Foundation of China (Grant no. 62306317), the Natural Science Foundation of Liaoning Provincial Education Department (Grant no. LJ212510165019), and Doctoral Scientific Research Start-up Foundation from Liaoning Normal University (Grant no. 2025BSL026)

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